Band-1 Hadamard Order 7444 — Miyamoto (Paley-II Seed H_1860 q=929 + C_1862) + Paley-II GF(61^2) IN SPK; Gap 2052 Still OPEN

H_1860 Paley-II PASS; C_1862 PASS; H_7444 Miyamoto PASS; H_7444 Paley-II GF(61^2) PASS → IN SPK via Paley-II q=3721; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7396_Miyamoto`; `2026-09-16_Band1_Hadamard_7156_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7444_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1860 and 7444 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1860} = Paley-II($q = 929$) and symmetric Paley conference C_{1862} over GF(1861) ($q = 1861 \equiv 1 \pmod 4$), **and** classical Paley-II H_{7444} = Paley-II($q = 3721 = 61^2$) over GF(61^2). Classical SPK for 7444: Paley-I **FAIL**s ($q = 7443 = 3^2 \cdot 827$ composite); Paley-II **PASS**es ($q = 3721 = 61^2 \equiv 1 \pmod 4$); Kronecker **FAIL**s ($4 \times 1861$). Therefore order **7444 is IN SPK via Paley-II** — same honesty class as 7396 / 6724 / 6436 (Miyamoto + IN SPK). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7396 Miyamoto+Paley-II IN SPK**. Prior SPK seeds **1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **1860** and **7444** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1860** and **7444**.

- H_{1860} = Paley-II ($q = 929$) — preferred seed
- C_{1862} = Paley conference over GF(1861), $q = 1861$
- H_{7444} = Miyamoto(H_{1860}, C_{1862})
- H_{7444} = Paley-II ($q = 3721 = 61^2$) over GF(61^2) — classical SPK path
- Paley-I for 7444: **SKIP** ($q = 7443$ composite)
- Kronecker for 7444: **SKIP** ($4 \cdot 1861$ not admissible)

SPK: 7444 IN SPK via Paley-II $q = 3721 = 61^2$. Seeds 1860 and 7444 enter SPK. Prior SPK **1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7444.py`

Results. overall_pass=true; units 115/115; wall 665.2448s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7444 = 4 \cdot 1861$; 1861 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1860} and C_{1862} (Paley conference over prime field $q = 1861$).
- Seed: Paley-II $q = 929$ (prime $\equiv 1 \pmod 4$) → H_{1860} (Paley-I $q = 1859$ FAIL; Double FAIL).
- SPK: Paley-I $q = 7443$ **FAIL** (composite $3^2 \cdot 827$); Paley-II $q = 3721 = 61^2$ **PASS** (prime power $\equiv 1 \pmod 4$ over GF(61^2)); Kronecker **FAIL → IN SPK via Paley-II**.
- Prior: 7396 Miyamoto+Paley-II IN SPK shipped; this ship $q = 1861 \rightarrow n = 7444$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1860 Paley-II	Paley-II q=929	0	3.3826
C_1862	Paley conf GF(1861) q=1861	0	4.469
H_7444 Miyamoto	Miyamoto(H_1860 Paley-II, C_1862)	0	323.1698

Object	Method	max|HH^T-nI|	Wall (s)
H_7444 Paley-II	Paley-II q=3721=61^2 GF(61^2) (classical SPK)	0	326.9014
H_7444 Paley-I	SKIP q=7443=$3^2 \cdot 827$	—	—
H_7444 Kronecker	SKIP $4 \cdot 1861$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **7396** lineage (**1848** Paley-I DIRECT; **7396** Paley-II; **1788** Paley-I; **888/1776**; **860/1720/6884**; **1708** Paley-II; **6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1860** (Paley-II) + verified **7444** (Paley-II GF(61^2)); **exclude** Miyamoto-only bonuses. Known count 1470. Next constructive gap **2052** (OPEN).

Quantity	Value
7444 in SPK	**True** (Paley-II q=3721=61^2)
1860 in SPK	**True** (Paley-II q=929)
7396 / 1848 in SPK	**True** (prior)
7156 / 7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite constructions of 1860+7444 ONLY.

Repro

Lab: `/workspace/experiments/band1/hadamard_7444_miyamoto/`
Artifacts: `H1860_paleyII_pm1.npz`, `C1862_pm1.npz`, `H7444_miyamoto_pm1.npz`, `H7444_paleyII_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.